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REMOTE USER PLANE DEPLOYMENT AND CONFIGURATION

Abstract

Aspects of the subject disclosure may include, for example, a method that includes providing, by a processing system including a processor, a controller function for a user plane function (UPF) of a communication network; the controller function facilitates automated procedures for authentication, deployment, configuration, testing, and/or controlling availability of the UPF, independent of a source of the UPF. The method also includes providing, by the processing system, an interface to facilitate communication between the controller function and the UPF; the controller function uses the interface to facilitate the procedures. Other embodiments are disclosed.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/422,528, filed on Jan. 25, 2024, which is a continuation of and claims priority to U.S. patent application Ser. No. 17/097,879, filed Nov. 13, 2020, now U.S. Pat. No. 11,924,662. All sections of the aforementioned applications and patent are incorporated herein by reference in their entirety.

FIELD OF THE DISCLOSURE

[0002] The subject disclosure relates to wireless communication networks, and more particularly to deployment and configuration of user plane functions (UPFs) in 5G core networks.

BACKGROUND

[0003] Communication networks, particularly 5G core networks, typically include a user plane controller and user plane functions (UPFs). The 5G core networks use UPFs from a variety of vendors. Currently, different vendors use proprietary methods and interfaces to configure, test, deploy and control their respective UPFs. This results in a complex network with multiple vendor-specific implementations. FIG. 2A-1 is a schematic illustration **800** of an enterprise **1103** using UPFs **1104** from multiple vendors **1102**, with each vendor using proprietary configuration and control interfaces for their respective UPFs. As is understood in the art, each vendor uses a proprietary extension to the N4 interface between the control plane and UPF, which requires a session management function (SMF) **1105** that supports the extensions. This results in a network that is difficult to manage and scale, with inefficient use of control plane resources.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

[0005] FIG. 1 is a block diagram illustrating an exemplary, non-limiting embodiment of a communications network in accordance with various aspects described herein.

[0006] FIG. 2A-1 schematically illustrates an enterprise using UPFs from multiple vendors, with each vendor using proprietary configuration and control interfaces for their respective UPFs.

[0007] FIG. 2A-2 schematically illustrates a remote UPF deployment and control function (RUDCF) for use with UPFs from multiple providers, in accordance with embodiments of the disclosure.

[0008] FIG. 2A-3 schematically illustrates capabilities of a RUDCF according to embodiments of the disclosure.

[0009] FIG. 2B-1 schematically illustrates logical functions of a RUDCF according to an embodiment of the disclosure.

[0010] FIG. 2B-2 schematically illustrates a functional architecture of an UPF according to an embodiment of the disclosure.

[0011] FIG. 2B-3 schematically illustrates an interface between a UPF and a RUDCF according to an embodiment of the disclosure.

[0012] FIG. 2C schematically illustrates a procedure for discovery by a session management

function (SMF) of a UPF controlled by an RUDCF, in accordance with embodiments of the disclosure.

[0013] FIG. 2D schematically illustrates an architecture for networking between an enterprise UPF and a service provider network, in accordance with embodiments of the disclosure.

[0014] FIG. 2E-1 schematically illustrates an architecture for implementation of an RUDCF to configure an enterprise UPF, according to an embodiment of the disclosure.

[0015] FIG. 2E-2 is a flowchart schematically illustrating a method for UPF configuration, according to an embodiment of the disclosure.

[0016] FIG. 2F-1 schematically illustrates an architecture for implementation of an RUDCF to perform a simulated test of an enterprise UPF configuration, according to an embodiment of the disclosure.

[0017] FIG. 2F-2 is a flowchart schematically illustrating a method for a UPF simulated test, according to an embodiment of the disclosure.

[0018] FIG. 2G-1 schematically illustrates an architecture for implementation of an RUDCF to perform a live test of an enterprise UPF configuration, according to an embodiment of the disclosure.

[0019] FIGS. 2G-2 and 2G-3 are connected flowcharts schematically illustrating a method for a UPF live test, according to an embodiment of the disclosure.

[0020] FIG. 2H-1 schematically illustrates implementation of a RUDCF with a UPF placed in service, according to an embodiment of the disclosure.

[0021] FIG. 2H-2 is a flowchart schematically illustrating a method for placing a UPF in service, according to an embodiment of the disclosure.

[0022] FIG. 3 is a block diagram illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein.

[0023] FIG. 4 is a block diagram of an example, non-limiting embodiment of a computing environment in accordance with various aspects described herein.

[0024] FIG. 5 is a block diagram of an example, non-limiting embodiment of a mobile network platform in accordance with various aspects described herein.

[0025] FIG. 6 is a block diagram of an example, non-limiting embodiment of a communication device in accordance with various aspects described herein.

DETAILED DESCRIPTION

[0026] The subject disclosure describes, among other things, illustrative embodiments for a vendor-independent network function that automates the deployment and control of enterprise or service provider UPFs. Other embodiments are described in the subject disclosure.

[0027] One or more aspects of the subject disclosure include a method that includes providing, by a processing system including a processor, a controller function for a user plane function (UPF) of a communication network; the controller function facilitates automated procedures for authentication, deployment, configuration, testing, and/or controlling availability of the UPF, independent of a source of the UPF. The method also includes providing, by the processing system, an interface to facilitate communication between the controller function and the UPF; the controller function uses the interface to facilitate the procedures.

[0028] One or more aspects of the subject disclosure include a device that includes a processing system including a processor and a memory that stores instructions; the instructions, when executed by the processing system, facilitate performance of operations. The operations include providing a controller function for a user plane function (UPF) of a communication network; the controller function facilitates automated procedures for authentication, deployment, configuration, testing, and/or controlling availability of the UPF, independent of a source of the UPF. The operations also include providing a service-based interface to facilitate communication between the controller function and the UPF; the controller function uses the interface to facilitate the procedures.

[0029] One or more aspects of the subject disclosure include a machine-readable medium comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations. The operations include providing a controller function for a user plane function (UPF) of a communication network; the controller function facilitates automated procedures for authentication, deployment, configuration, testing, and/or controlling availability of the UPF, independent of a source of the UPF, and the testing includes simulated testing and live network testing of a configuration of the UPF. The operations also include providing an interface to facilitate communication between the controller function and the UPF, wherein the controller function uses the interface to facilitate the procedures.

[0030] Referring now to FIG. 1, a block diagram is shown illustrating an example, non-limiting embodiment of a system **100** in accordance with various aspects described herein. For example, system **100** can facilitate in whole or in part providing a controller function for a user plane function (UPF) of a communication network; the controller function can facilitate automated procedures for authentication, deployment, configuration, testing, and/or controlling availability of the UPF, independent of a source of the UPF. In particular, a communications network **125** is presented for providing broadband access **110** to a plurality of data terminals **114** via access terminal **112**, wireless access **120** to a plurality of mobile devices **124** and vehicle **126** via base station or access point **122**, voice access **130** to a plurality of telephony devices **134**, via switching device **132** and/or media access **140** to a plurality of audio/video display devices **144** via media terminal **142**. In addition, communication network **125** is coupled to one or more content sources **175** of audio, video, graphics, text and/or other media. While broadband access **110**, wireless access **120**, voice access **130** and media access **140** are shown separately, one or more of these forms of access can be combined to provide multiple access services to a single client device (e.g., mobile devices **124** can receive media content via media terminal **142**, data terminal **114** can be provided voice access via switching device **132**, and so on).

[0031] The communications network **125** includes a plurality of network elements (NE) **150**, **152**, **154**, **156**, etc. for facilitating the broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or the distribution of content from content sources **175**. The communications network **125** can include a circuit switched or packet switched network, a voice over Internet protocol (VoIP) network, Internet protocol (IP) network, a cable network, a passive or active optical network, a 4G, 5G, or higher generation wireless access network, WIMAX network, UltraWideband network, personal area network or other wireless access network, a broadcast satellite network and/or other communications network.

[0032] In various embodiments, the access terminal **112** can include a digital subscriber line access multiplexer (DSLAM), cable modem termination system (CMTS), optical line terminal (OLT) and/or other access terminal. The data terminals **114** can include personal computers, laptop computers, netbook computers, tablets or other computing devices along with digital subscriber line (DSL) modems, data over coax service interface specification (DOCSIS) modems or other cable modems, a wireless modem such as a 4G, 5G, or higher generation modem, an optical modem and/or other access devices.

[0033] In various embodiments, the base station or access point **122** can include a 4G, 5G, or higher generation base station, an access point that operates via an 802.11 standard such as 802.11n, 802.11ac or other wireless access terminal. The mobile devices **124** can include mobile phones, e-readers, tablets, phablets, wireless modems, and/or other mobile computing devices.

[0034] In various embodiments, the switching device **132** can include a private branch exchange or central office switch, a media services gateway, VoIP gateway or other gateway device and/or other switching device. The telephony devices **134** can include traditional telephones (with or without a terminal adapter), VoIP telephones and/or other telephony devices.

[0035] In various embodiments, the media terminal **142** can include a cable head-end or other TV head-end, a satellite receiver, gateway or other media terminal **142**. The display devices **144** can

include televisions with or without a set top box, personal computers and/or other display devices. [0036] In various embodiments, the content sources **175** include broadcast television and radio sources, video on demand platforms and streaming video and audio services platforms, one or more content data networks, data servers, web servers and other content servers, and/or other sources of media.

[0037] In various embodiments, the communications network **125** can include wired, optical and/or wireless links and the network elements **150, 152, 154, 156**, etc. can include service switching points, signal transfer points, service control points, network gateways, media distribution hubs, servers, firewalls, routers, edge devices, switches and other network nodes for routing and controlling communications traffic over wired, optical and wireless links as part of the Internet and other public networks as well as one or more private networks, for managing subscriber access, for billing and network management and for supporting other network functions.

[0038] FIG. 2A-2 is a schematic illustration **900** of a remote UPF deployment and control function (RUDCF) for use with UPFs from multiple vendors, in accordance with embodiments of the disclosure. As shown in FIG. 2A-2, enterprises **1205** (customers of service provider **1201**) use UPFs **1211-1215** from different vendors. Service provider **1201** includes RUDCF **1210**, which standardizes and automates control of the various UPFs. In an embodiment, the service provider also provides a session management function (SMF) **1250** that can interoperate with the various enterprise UPFs.

[0039] FIG. 2A-3 is a schematic illustration **1000** of control of a UPF **1301** by a RUDCF **1310**. According to embodiments of the disclosure, the RUDCF enables a large number of UPFs from multiple vendors to be configured, deployed and controlled using a single standardized network function. In this embodiment, RUDCF **1310** and UPF **1301** communicate via a service-based interface (SBI) **1315**.

[0040] FIG. 2B-1 is a schematic illustration **1200** of a functional architecture of a RUDCF according to an embodiment of the disclosure. RUDCF **2100** includes a control function **2101** which orchestrates UPF authentication, configuration, testing, and service state (i.e., “available” state and “not available” state). UPF authentication function **2102** provides application layer authentication, in addition to other authentication methods described further herein. UPF platform configuration function **2103** configures UPFs and updates UPF configurations. Simulated network test function (STF) **2104** tests the UPF configuration using a simulated SMF (with an N4 interface) and gNB (with an N3 interface). Live network test function **2105** tests UPF configurations using a production network.

[0041] FIG. 2B-2 is a schematic illustration **1300** of a functional architecture of a UPF according to an embodiment of the disclosure. As shown in FIG. 2B-2, UPF **1301** includes a user plane control function (UPCF) **1302**. The UPCF receives and executes commands from the RUDCF regarding authentication, configuration, testing and service state.

[0042] FIG. 2B-3 is a schematic illustration **1400** of an interface between a UPF and a RUDCF according to an embodiment of the disclosure. In this embodiment, interface **2301** is implemented using Third Generation Partnership Project (3GPP) Service Based Architecture (SBA) concepts. Interface **2301** is a service based interface (SBI) with services **2310** defined at the RUDCF and services **2320** defined at the UPF. The SBI **2301** may also be referred to as a reference point defined between the RUDCF and the UPF.

[0043] Services **2310** provided at the RUDCF can include authenticating the UPF, receiving an existing UPF configuration, providing a new UPF configuration, providing notification of changes in a UPF configuration, receiving UPF test results, receiving a UPF status, etc. Services **2320** provided at the UPF can include test control (for a simulated test or a live test), changing a service state in a Network Repository Function (NRF) between a “registered in NRF” state (i.e. when the UPF is in service) and a “not registered” state (i.e. when the UPF is out of service), providing notification of changes in a UPF configuration, etc.

[0044] In various embodiments, a UPF may be operated either by an enterprise or a service provider. At its initial configuration, a UPF requests and receives a configuration from the RUDCF. After the UPF is configured, the RUDCF sends control instructions to the UPF. The RUDCF may use a user data repository (UDR) to store UPF configuration data.

[0045] FIG. 2C is a schematic illustration **2200** of a procedure for discovery by a session management function (SMF) of a UPF controlled by an RUDCF, in accordance with embodiments of the disclosure. A UPF **3101** is discoverable by one or more SMFs **3102** when the UPF is placed in service, and is not discoverable when removed from service. When a RUDCF **3103** instructs the UPF to go into (or out of) service, the UPF registers (or de-registers) at the NRF **3104**.

[0046] The SMFs use the NRF to discover UPFs managed by the RUDCF. In an embodiment, one or more SMF serving areas **3105** are defined to serve UPFs controlled by the RUDCF. In this embodiment, UPFs controlled by the RUDCF register at the NRF with a new service name, e.g. “enterprise”. An access management function (AMF) in the enterprise serving area can subscribe to NRF events for UPFs with this service name; when a UPF is added or removed, the SMFs in the SMF serving area are notified by the NRF.

[0047] In an embodiment, an AMF subscribes to events for UPFs that have the service name “enterprise” (step **3111**). A decision to put a UPF in service (step **3112**) results in the RUDCF sending an instruction to the UPF (step **3113**) to register at the NRF. The UPF registers (step **3114**), resulting in its state changing to “registered” with a new service name (e.g., in this embodiment “enterprise”). The NRF notifies the SMFs of the new UPF registration and service name (step **3115**). Associations between the SMFs and the UPF may then be established (step **3116**) according to the Packet Forwarding Control Protocol (PFCP).

[0048] An example of implementation of an RUDCF, in which a new UPF is configured, tested and placed in service, is detailed below.

[0049] FIG. 2D is a schematic illustration **2700** of an architecture for networking between an enterprise UPF and a service provider network, in accordance with embodiments of the disclosure. In an embodiment, an enterprise UPF **4101** communicates with a service provider **4110** via data network **4102** (which may be a public or private network). The service provider includes a data network **4112** providing access to a Radio Access Network (RAN) **4113**, control plane functions **4114**, and network security functions **4115**.

[0050] FIG. 2E-1 is a schematic illustration **2900** of an architecture for implementation of an RUDCF to configure an enterprise UPF, according to an embodiment of the disclosure. UPF **5101** communicates with a data network **5112** of service provider **5110**. Communications between the UPF **5101** and RUDCF **5120** of the service provider are routed through the network security function **5113**. In this embodiment, the UPF discovers the RUDCF using NRF **5124**. The control function **5121** of the RUDCF and the UPF **5101** perform a mutual authentication and authorization procedure; in an embodiment, this may be done using 3GPP standard SBI security procedures. The UPF obtains its configuration from the RUDCF; configuration data is stored in UDR **5126**.

[0051] FIG. 2E-2 is a flowchart **3200** schematically illustrating a method for UPF configuration, according to an embodiment of the disclosure. In this embodiment, the UPF discovers the NRF (step **5202**), and the UPF and NRF perform a mutual authentication. The UPF then registers with the NRF (step **5204**). The UPF then discovers the RUDCF; the UPF and RUDCF perform a mutual authentication (step **5206**). The authorization function **5127** of the RUDCF provides a base configuration for the UPF (step **5208**); the UPF then informs the configuration function **5129** of the RUDCF (step **5210**). The RUDCF configuration function generates a configuration for the UPF (step **5212**), which is stored in the UDR **5126**. The UPF then subscribes to configuration changes at the RUDCF (step **5214**), and the RUDCF configuration function subscribes to configuration changes at the UPF (step **5216**).

[0052] FIG. 2F-1 is a schematic illustration **3600** of an architecture for implementation of an RUDCF to perform a simulated test of an enterprise UPF configuration, according to an

embodiment of the disclosure. In this embodiment, enterprise **6101** includes a data network **6112** in communication with UPF **6110** and a simulated host **6114**. The UPF **6110** communicates with a data network **6122** of service provider **6120**. Communications between the UPF **6110** and RUDCF **6130** of the service provider are routed through the network security function **6123**. In this embodiment, an association is established between the simulated test function (STF) **6134** of the service provider and the UPF, using a N4 interface implemented according to the PFCP protocol. Downlink (DL) and uplink (UL) traffic is enabled between the STF, the UPF and the simulated host using a N3 interface.

[0053] FIG. 2F-2 is a flowchart **3800** schematically illustrating a method for a UPF simulated test, according to an embodiment of the disclosure. The RUDCF instructs the UPF to start a simulated test (step **6202**); a PFCP association is established between the STF of the RUDCF and the UPF (step **6204**). One or more user plane sessions are established for the simulated test (step **6206**); uplink and downlink data is exchanged between the STF, the UPF, and the simulated host (step **6208**). The test (step **6210**) includes testing each PFCP session rule, e.g. packet detection rules (PDR), forwarding action rules (FAR), usage reporting rules (URR), quality of service (QoS) enforcement rules (QER), and buffering action rules (BAR). UPF usage reporting is also performed (step **6212**). To conclude the test, the user plane sessions are released (step **6214**), and the UPF sends test data to the RUDCF (step **6216**). The UPF then initiates a PFCP association release (step **6218**) to test a UPF shutdown (alternatively, detaching of the UPF from the network). The test results are then evaluated (step **6220**).

[0054] In additional embodiments, the simulated test function (STF) of the RUDCF can be used after the UPF is deployed, e.g. for troubleshooting, testing of new UPF configurations, and/or testing of new UPF functions.

[0055] FIG. 2G-1 is a schematic illustration **4300** of an architecture for implementation of an RUDCF to perform a live test of an enterprise UPF configuration, according to an embodiment of the disclosure. In this embodiment, the objective is to perform an end-to-end test of the UPF in a production network, including UPF registration with the NRF and UPF discovery by the SMF(s) using the NRF.

[0056] In this embodiment, the service provider **7102** includes RUDCF **7120** that orchestrates testing the enterprise UPF **7101**, using a test user equipment (test UE) **7140**; enterprise **7100** includes a data network, the UPF **7101** and a simulated host **7103**. The test UE has a packet data unit (PDU) session on two data network names (DNNs): a target DNN **7143** for testing the enterprise UPF, and a live test DNN **7141** for test instructions from the RUDCF and for test results reported to the live test function (LTF) **7121** of the RUDCF. The service provider also includes an access and mobility management function (AMF) **7130** in communication with the test UE **7140**. The live test DNN uses a UPF **7125** dedicated to the live test function. The test UE **7140** and the UPF under test **7101** report test results to the RUDCF; the RUDCF evaluates the test results.

[0057] FIGS. 2G-2 and 2G-3 are connected flowcharts **4400**, **4700** schematically illustrating a method for a UPF live test, according to an embodiment of the disclosure. The SMF subscribes to all the UPFs in the enterprise SMF serving area (step **7202**); the UE live test control function registers with the RUDCF (step **7204**). The RUDCF initiates the test (step **7206**); the UPF then registers with the NRF (step **7208**). The NRF informs the SMF(s) of the UPF registration (step **7210**). A PFCP association is established between the SMF and the UPF under test (step **7212**). The live test function (LTF) of the RUDCF sends instructions to the test UE (step **7214**) to begin the live test. A PDU session is requested on the target DNN, and the SMF selects the UPF for the DNN (step **7216**). PFCP user plane sessions are then established between the SMF and the enterprise UPF (step **7218**).

[0058] Uplink and downlink data is exchanged between the test UE and the enterprise UPF, and between the enterprise UPF and the simulated host (step **7220**); UPF usage reporting is also performed (step **7222**). Test data is sent to the RUDCF (step **7224**) from the UPF under test and the

test UE. When the RUDCF determines that the test is complete (step **7226**), it instructs the UE to end the test; the test UE then sends the final test data (step **7228**), and the user plane sessions between the SMF and UPF are released (step **7230**). The RUDCF ends the live test (step **7232**), and the UPF de-registers at the NRF (step **7234**). The PFCP associations between the SMF and UPF are released (step **7236**), and the final test data is sent from the UPF to the RUDCF (step **7238**).

[0059] After live testing is completed successfully, the UPF may be allowed to go into service.

FIG. **2H-1** is a schematic illustration **5100** of implementation of a RUDCF with a UPF placed in service, according to an embodiment of the disclosure. Enterprise **8100** includes a data network and enterprise UPF **8101**, and communicates with service provider **8102**; the service provider includes RUDCF **8120**, SMF(s) **8110** and NRF **8105**. When instructed by the RUDCF to go into service, the UPF registers at the NRF and changes its state to “registered;” the UPF then is discoverable by the SMF(s).

[0060] FIG. **2H-2** is a flowchart **5200** schematically illustrating a method for placing a UPF in service, according to an embodiment of the disclosure. The RUDCF instructs the UPF to go into service (step **8202**); the UPF then registers with the NRF (step **8204**). The NRF informs the SMF(s) that the UPF is available (step **8206**). PFCP associations are established (step **8208**) between the UPF and each SMF in the SMF serving area. The RUDCF then confirms that the UPF is in service (step **8210**).

[0061] While for purposes of simplicity of explanation, the respective processes are shown and described as a series of blocks in FIGS. **2E-2**, **2F-2**, **2G-2**, **2G-3** and **2H2**, it is to be understood and appreciated that the claimed subject matter is not limited by the order of the blocks, as some blocks may occur in different orders and/or concurrently with other blocks from what is depicted and described herein. Moreover, not all illustrated blocks may be required to implement the methods described herein.

[0062] Referring now to FIG. **3**, a block diagram **300** is shown illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein. In particular a virtualized communication network is presented that can be used to implement some or all of the subsystems and functions of system **100**, the subsystems and functions of the systems shown in FIGS. **2A-2**, **2A-3**, **2B-1**, **2B-2**, **2B-3**, **2C**, **2D**, **2E-1**, **2F-1**, **2G-1** and **2H-1**, and the methods presented in FIGS. **2E-2**, **2F-2**, **2G-2**, **2G-3** and **2H-2**. For example, virtualized communication network **300** can facilitate in whole or in part providing a controller function for a user plane function (UPF) of a communication network, where the controller function facilitates automated procedures for authentication, deployment, configuration, testing, and/or controlling availability of the UPF, independent of a source of the UPF; and providing an interface to facilitate communication between the controller function and the UPF, where the controller function uses the interface to facilitate the procedures.

[0063] In particular, a cloud networking architecture is shown that leverages cloud technologies and supports rapid innovation and scalability via a transport layer **350**, a virtualized network function cloud **325** and/or one or more cloud computing environments **375**. In various embodiments, this cloud networking architecture is an open architecture that leverages application programming interfaces (APIs); reduces complexity from services and operations; supports more nimble business models; and rapidly and seamlessly scales to meet evolving customer requirements including traffic growth, diversity of traffic types, and diversity of performance and reliability expectations.

[0064] In contrast to traditional network elements-which are typically integrated to perform a single function, the virtualized communication network employs virtual network elements (VNEs) **330**, **332**, **334**, etc. that perform some or all of the functions of network elements **150**, **152**, **154**, **156**, etc. For example, the network architecture can provide a substrate of networking capability, often called Network Function Virtualization Infrastructure (NFVI) or simply infrastructure that is capable of being directed with software and Software Defined Networking (SDN) protocols to

perform a broad variety of network functions and services. This infrastructure can include several types of substrates. The most typical type of substrate being servers that support Network Function Virtualization (NFV), followed by packet forwarding capabilities based on generic computing resources, with specialized network technologies brought to bear when general purpose processors or general purpose integrated circuit devices offered by merchants (referred to herein as merchant silicon) are not appropriate. In this case, communication services can be implemented as cloud-centric workloads.

[0065] As an example, a traditional network element **150** (shown in FIG. **1**), such as an edge router can be implemented via a VNE **330** composed of NFV software modules, merchant silicon, and associated controllers. The software can be written so that increasing workload consumes incremental resources from a common resource pool, and moreover so that it's elastic: so the resources are only consumed when needed. In a similar fashion, other network elements such as other routers, switches, edge caches, and middle-boxes are instantiated from the common resource pool. Such sharing of infrastructure across a broad set of uses makes planning and growing infrastructure easier to manage.

[0066] In an embodiment, the transport layer **350** includes fiber, cable, wired and/or wireless transport elements, network elements and interfaces to provide broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or access to content sources **175** for distribution of content to any or all of the access technologies. In particular, in some cases a network element needs to be positioned at a specific place, and this allows for less sharing of common infrastructure. Other times, the network elements have specific physical layer adapters that cannot be abstracted or virtualized, and might require special DSP code and analog front-ends (AFEs) that do not lend themselves to implementation as VNEs **330**, **332** or **334**. These network elements can be included in transport layer **350**.

[0067] The virtualized network function cloud **325** interfaces with the transport layer **350** to provide the VNEs **330**, **332**, **334**, etc. to provide specific NFVs. In particular, the virtualized network function cloud **325** leverages cloud operations, applications, and architectures to support networking workloads. The virtualized network elements **330**, **332** and **334** can employ network function software that provides either a one-for-one mapping of traditional network element function or alternately some combination of network functions designed for cloud computing. For example, VNEs **330**, **332** and **334** can include route reflectors, domain name system (DNS) servers, and dynamic host configuration protocol (DHCP) servers, system architecture evolution (SAE) and/or mobility management entity (MME) gateways, broadband network gateways, IP edge routers for IP-VPN, Ethernet and other services, load balancers, distributors and other network elements. Because these elements don't typically need to forward large amounts of traffic, their workload can be distributed across a number of servers-each of which adds a portion of the capability, and overall which creates an elastic function with higher availability than its former monolithic version. These virtual network elements **330**, **332**, **334**, etc. can be instantiated and managed using an orchestration approach similar to those used in cloud compute services.

[0068] The cloud computing environments **375** can interface with the virtualized network function cloud **325** via APIs that expose functional capabilities of the VNEs **330**, **332**, **334**, etc. to provide the flexible and expanded capabilities to the virtualized network function cloud **325**. In particular, network workloads may have applications distributed across the virtualized network function cloud **325** and cloud computing environment **375** and in the commercial cloud, or might simply orchestrate workloads supported entirely in NFV infrastructure from these third party locations.

[0069] Turning now to FIG. **4**, there is illustrated a block diagram of a computing environment in accordance with various aspects described herein. In order to provide additional context for various embodiments of the embodiments described herein, FIG. **4** and the following discussion are intended to provide a brief, general description of a suitable computing environment **400** in which the various embodiments of the subject disclosure can be implemented. In particular, computing

environment **400** can be used in the implementation of network elements **150, 152, 154, 156**, access terminal **112**, base station or access point **122**, switching device **132**, media terminal **142**, and/or VNEs **330, 332, 334**, etc. Each of these devices can be implemented via computer-executable instructions that can run on one or more computers, and/or in combination with other program modules and/or as a combination of hardware and software. For example, computing environment **400** can facilitate in whole or in part providing a controller function for a user plane function (UPF) of a communication network, where the controller function facilitates automated procedures for authentication, deployment, configuration, testing, and/or controlling availability of the UPF, independent of a source of the UPF; and providing a service-based interface to facilitate communication between the controller function and the UPF, where the controller function uses the interface to facilitate the procedures.

[0070] Generally, program modules comprise routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the methods can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, minicomputers, mainframe computers, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

[0071] As used herein, a processing circuit includes one or more processors as well as other application specific circuits such as an application specific integrated circuit, digital logic circuit, state machine, programmable gate array or other circuit that processes input signals or data and that produces output signals or data in response thereto. It should be noted that while any functions and features described herein in association with the operation of a processor could likewise be performed by a processing circuit.

[0072] The illustrated embodiments of the embodiments herein can be also practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0073] Computing devices typically comprise a variety of media, which can comprise computer-readable storage media and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media can be any available storage media that can be accessed by the computer and comprises both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable instructions, program modules, structured data or unstructured data.

[0074] Computer-readable storage media can comprise, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, compact disk read only memory (CD-ROM), digital versatile disk (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

[0075] Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

[0076] Communications media typically embody computer-readable instructions, data structures,

program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and comprises any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media comprise wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.

[0077] With reference again to FIG. 4, the example environment can comprise a computer **402**, the computer **402** comprising a processing unit **404**, a system memory **406** and a system bus **408**. The system bus **408** couples system components including, but not limited to, the system memory **406** to the processing unit **404**. The processing unit **404** can be any of various commercially available processors. Dual microprocessors and other multiprocessor architectures can also be employed as the processing unit **404**.

[0078] The system bus **408** can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures. The system memory **406** comprises ROM **410** and RAM **412**. A basic input/output system (BIOS) can be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, which BIOS contains the basic routines that help to transfer information between elements within the computer **402**, such as during startup. The RAM **412** can also comprise a high-speed RAM such as static RAM for caching data.

[0079] The computer **402** further comprises an internal hard disk drive (HDD) **414** (e.g., EIDE, SATA), which internal HDD **414** can also be configured for external use in a suitable chassis (not shown), a magnetic floppy disk drive (FDD) **416**, (e.g., to read from or write to a removable diskette **418**) and an optical disk drive **420**, (e.g., reading a CD-ROM disk **422** or, to read from or write to other high capacity optical media such as the DVD). The HDD **414**, magnetic FDD **416** and optical disk drive **420** can be connected to the system bus **408** by a hard disk drive interface **424**, a magnetic disk drive interface **426** and an optical drive interface **428**, respectively. The hard disk drive interface **424** for external drive implementations comprises at least one or both of Universal Serial Bus (USB) and Institute of Electrical and Electronics Engineers (IEEE) 1394 interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.

[0080] The drives and their associated computer-readable storage media provide nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For the computer **402**, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to a hard disk drive (HDD), a removable magnetic diskette, and a removable optical media such as a CD or DVD, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, such as zip drives, magnetic cassettes, flash memory cards, cartridges, and the like, can also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.

[0081] A number of program modules can be stored in the drives and RAM **412**, comprising an operating system **430**, one or more application programs **432**, other program modules **434** and program data **436**. All or portions of the operating system, applications, modules, and/or data can also be cached in the RAM **412**. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.

[0082] A user can enter commands and information into the computer **402** through one or more wired/wireless input devices, e.g., a keyboard **438** and a pointing device, such as a mouse **440**. Other input devices (not shown) can comprise a microphone, an infrared (IR) remote control, a joystick, a game pad, a stylus pen, touch screen or the like. These and other input devices are often

connected to the processing unit **404** through an input device interface **442** that can be coupled to the system bus **408**, but can be connected by other interfaces, such as a parallel port, an IEEE 1394 serial port, a game port, a universal serial bus (USB) port, an IR interface, etc.

[0083] A monitor **444** or other type of display device can be also connected to the system bus **408** via an interface, such as a video adapter **446**. It will also be appreciated that in alternative embodiments, a monitor **444** can also be any display device (e.g., another computer having a display, a smart phone, a tablet computer, etc.) for receiving display information associated with computer **402** via any communication means, including via the Internet and cloud-based networks. In addition to the monitor **444**, a computer typically comprises other peripheral output devices (not shown), such as speakers, printers, etc.

[0084] The computer **402** can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote computer(s) **448**. The remote computer(s) **448** can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically comprises many or all of the elements described relative to the computer **402**, although, for purposes of brevity, only a remote memory/storage device **450** is illustrated. The logical connections depicted comprise wired/wireless connectivity to a local area network (LAN) **452** and/or larger networks, e.g., a wide area network (WAN) **454**. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the Internet.

[0085] When used in a LAN networking environment, the computer **402** can be connected to the LAN **452** through a wired and/or wireless communication network interface or adapter **456**. The adapter **456** can facilitate wired or wireless communication to the LAN **452**, which can also comprise a wireless AP disposed thereon for communicating with the adapter **456**.

[0086] When used in a WAN networking environment, the computer **402** can comprise a modem **458** or can be connected to a communications server on the WAN **454** or has other means for establishing communications over the WAN **454**, such as by way of the Internet. The modem **458**, which can be internal or external and a wired or wireless device, can be connected to the system bus **408** via the input device interface **442**. In a networked environment, program modules depicted relative to the computer **402** or portions thereof, can be stored in the remote memory/storage device **450**. It will be appreciated that the network connections shown are example and other means of establishing a communications link between the computers can be used.

[0087] The computer **402** can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, restroom), and telephone. This can comprise Wireless Fidelity (Wi-Fi) and BLUETOOTH® wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

[0088] Wi-Fi can allow connection to the Internet from a couch at home, a bed in a hotel room or a conference room at work, without wires. Wi-Fi is a wireless technology similar to that used in a cell phone that enables such devices, e.g., computers, to send and receive data indoors and out; anywhere within the range of a base station. Wi-Fi networks use radio technologies called IEEE 802.11 (a, b, g, n, ac, ag, etc.) to provide secure, reliable, fast wireless connectivity. A Wi-Fi network can be used to connect computers to each other, to the Internet, and to wired networks (which can use IEEE 802.3 or Ethernet). Wi-Fi networks operate in the unlicensed 2.4 and 5 GHz radio bands for example or with products that contain both bands (dual band), so the networks can provide real-world performance similar to the basic 10BaseT wired Ethernet networks used in many offices.

[0089] Turning now to FIG. 5, an embodiment 500 of a mobile network platform 510 is shown that is an example of network elements 150, 152, 154, 156, and/or VNEs 330, 332, 334, etc. For example, platform 510 can facilitate in whole or in part providing a controller function for a user plane function (UPF) of a communication network, where the controller function facilitates automated procedures for authentication, deployment, configuration, testing, and/or controlling availability of the UPF, independent of a source of the UPF, and where the testing includes simulated testing and live network testing of a configuration of the UPF; and providing an interface to facilitate communication between the controller function and the UPF, where the controller function uses the interface to facilitate the procedures. In one or more embodiments, the mobile network platform 510 can generate and receive signals transmitted and received by base stations or access points such as base station or access point 122. Generally, mobile network platform 510 can comprise components, e.g., nodes, gateways, interfaces, servers, or disparate platforms, that facilitate both packet-switched (PS) (e.g., internet protocol (IP), frame relay, asynchronous transfer mode (ATM)) and circuit-switched (CS) traffic (e.g., voice and data), as well as control generation for networked wireless telecommunication. As a non-limiting example, mobile network platform 510 can be included in telecommunications carrier networks, and can be considered carrier-side components as discussed elsewhere herein. Mobile network platform 510 comprises CS gateway node(s) 512 which can interface CS traffic received from legacy networks like telephony network(s) 540 (e.g., public switched telephone network (PSTN), or public land mobile network (PLMN)) or a signaling system #7 (SS7) network 560. CS gateway node(s) 512 can authorize and authenticate traffic (e.g., voice) arising from such networks. Additionally, CS gateway node(s) 512 can access mobility, or roaming, data generated through SS7 network 560; for instance, mobility data stored in a visited location register (VLR), which can reside in memory 530. Moreover, CS gateway node(s) 512 interfaces CS-based traffic and signaling and PS gateway node(s) 518. As an example, in a 3GPP UMTS network, CS gateway node(s) 512 can be realized at least in part in gateway GPRS support node(s) (GGSN). It should be appreciated that functionality and specific operation of CS gateway node(s) 512, PS gateway node(s) 518, and serving node(s) 516, is provided and dictated by radio technology(ies) utilized by mobile network platform 510 for telecommunication over a radio access network 520 with other devices, such as a radiotelephone 575.

[0090] In addition to receiving and processing CS-switched traffic and signaling, PS gateway node(s) 518 can authorize and authenticate PS-based data sessions with served mobile devices. Data sessions can comprise traffic, or content(s), exchanged with networks external to the mobile network platform 510, like wide area network(s) (WANs) 550, enterprise network(s) 570, and service network(s) 580, which can be embodied in local area network(s) (LANs), can also be interfaced with mobile network platform 510 through PS gateway node(s) 518. It is to be noted that WANs 550 and enterprise network(s) 570 can embody, at least in part, a service network(s) like IP multimedia subsystem (IMS). Based on radio technology layer(s) available in technology resource(s) or radio access network 520, PS gateway node(s) 518 can generate packet data protocol contexts when a data session is established; other data structures that facilitate routing of packetized data also can be generated. To that end, in an aspect, PS gateway node(s) 518 can comprise a tunnel interface (e.g., tunnel termination gateway (TTG) in 3GPP UMTS network(s) (not shown)) which can facilitate packetized communication with disparate wireless network(s), such as Wi-Fi networks.

[0091] In embodiment 500, mobile network platform 510 also comprises serving node(s) 516 that, based upon available radio technology layer(s) within technology resource(s) in the radio access network 520, convey the various packetized flows of data streams received through PS gateway node(s) 518. It is to be noted that for technology resource(s) that rely primarily on CS communication, server node(s) can deliver traffic without reliance on PS gateway node(s) 518; for example, server node(s) can embody at least in part a mobile switching center. As an example, in a

3GPP UMTS network, serving node(s) **516** can be embodied in serving GPRS support node(s) (SGSN).

[0092] For radio technologies that exploit packetized communication, server(s) **514** in mobile network platform **510** can execute numerous applications that can generate multiple disparate packetized data streams or flows, and manage (e.g., schedule, queue, format . . .) such flows. Such application(s) can comprise add-on features to standard services (for example, provisioning, billing, customer support . . .) provided by mobile network platform **510**. Data streams (e.g., content(s) that are part of a voice call or data session) can be conveyed to PS gateway node(s) **518** for authorization/authentication and initiation of a data session, and to serving node(s) **516** for communication thereafter. In addition to application server, server(s) **514** can comprise utility server(s), a utility server can comprise a provisioning server, an operations and maintenance server, a security server that can implement at least in part a certificate authority and firewalls as well as other security mechanisms, and the like. In an aspect, security server(s) secure communication served through mobile network platform **510** to ensure network's operation and data integrity in addition to authorization and authentication procedures that CS gateway node(s) **512** and PS gateway node(s) **518** can enact. Moreover, provisioning server(s) can provision services from external network(s) like networks operated by a disparate service provider; for instance, WAN **550** or Global Positioning System (GPS) network(s) (not shown). Provisioning server(s) can also provision coverage through networks associated to mobile network platform **510** (e.g., deployed and operated by the same service provider), such as the distributed antennas networks shown in FIG. **1**(s) that enhance wireless service coverage by providing more network coverage.

[0093] It is to be noted that server(s) **514** can comprise one or more processors configured to confer at least in part the functionality of mobile network platform **510**. To that end, the one or more processor can execute code instructions stored in memory **530**, for example. It is should be appreciated that server(s) **514** can comprise a content manager, which operates in substantially the same manner as described hereinbefore.

[0094] In example embodiment **500**, memory **530** can store information related to operation of mobile network platform **510**. Other operational information can comprise provisioning information of mobile devices served through mobile network platform **510**, subscriber databases; application intelligence, pricing schemes, e.g., promotional rates, flat-rate programs, couponing campaigns; technical specification(s) consistent with telecommunication protocols for operation of disparate radio, or wireless, technology layers; and so forth. Memory **530** can also store information from at least one of telephony network(s) **540**, WAN **550**, SS7 network **560**, or enterprise network(s) **570**. In an aspect, memory **530** can be, for example, accessed as part of a data store component or as a remotely connected memory store.

[0095] In order to provide a context for the various aspects of the disclosed subject matter, FIG. **5**, and the following discussion, are intended to provide a brief, general description of a suitable environment in which the various aspects of the disclosed subject matter can be implemented. While the subject matter has been described above in the general context of computer-executable instructions of a computer program that runs on a computer and/or computers, those skilled in the art will recognize that the disclosed subject matter also can be implemented in combination with other program modules. Generally, program modules comprise routines, programs, components, data structures, etc. that perform particular tasks and/or implement particular abstract data types.

[0096] Turning now to FIG. **6**, an illustrative embodiment of a communication device **600** is shown. The communication device **600** can serve as an illustrative embodiment of devices such as data terminals **114**, mobile devices **124**, vehicle **126**, display devices **144** or other client devices for communication via either communications network **125**. For example, computing device **600** can include an enterprise device with a UPF having features as described herein.

[0097] The communication device **600** can comprise a wireline and/or wireless transceiver **602** (herein transceiver **602**), a user interface (UI) **604**, a power supply **614**, a location receiver **616**, a

motion sensor **618**, an orientation sensor **620**, and a controller **606** for managing operations thereof. The transceiver **602** can support short-range or long-range wireless access technologies such as Bluetooth®, ZigBee®, WiFi, DECT, or cellular communication technologies, just to mention a few (Bluetooth® and ZigBee® are trademarks registered by the Bluetooth® Special Interest Group and the ZigBee® Alliance, respectively). Cellular technologies can include, for example, CDMA-1X, UMTS/HSDPA, GSM/GPRS, TDMA/EDGE, EV/DO, WiMAX, SDR, LTE, as well as other next generation wireless communication technologies as they arise. The transceiver **602** can also be adapted to support circuit-switched wireline access technologies (such as PSTN), packet-switched wireline access technologies (such as TCP/IP, VoIP, etc.), and combinations thereof.

[0098] The UI **604** can include a depressible or touch-sensitive keypad **608** with a navigation mechanism such as a roller ball, a joystick, a mouse, or a navigation disk for manipulating operations of the communication device **600**. The keypad **608** can be an integral part of a housing assembly of the communication device **600** or an independent device operably coupled thereto by a tethered wireline interface (such as a USB cable) or a wireless interface supporting for example Bluetooth®. The keypad **608** can represent a numeric keypad commonly used by phones, and/or a QWERTY keypad with alphanumeric keys. The UI **604** can further include a display **610** such as monochrome or color LCD (Liquid Crystal Display), OLED (Organic Light Emitting Diode) or other suitable display technology for conveying images to an end user of the communication device **600**. In an embodiment where the display **610** is touch-sensitive, a portion or all of the keypad **608** can be presented by way of the display **610** with navigation features.

[0099] The display **610** can use touch screen technology to also serve as a user interface for detecting user input. As a touch screen display, the communication device **600** can be adapted to present a user interface having graphical user interface (GUI) elements that can be selected by a user with a touch of a finger. The display **610** can be equipped with capacitive, resistive or other forms of sensing technology to detect how much surface area of a user's finger has been placed on a portion of the touch screen display. This sensing information can be used to control the manipulation of the GUI elements or other functions of the user interface. The display **610** can be an integral part of the housing assembly of the communication device **600** or an independent device communicatively coupled thereto by a tethered wireline interface (such as a cable) or a wireless interface.

[0100] The UI **604** can also include an audio system **612** that utilizes audio technology for conveying low volume audio (such as audio heard in proximity of a human ear) and high volume audio (such as speakerphone for hands free operation). The audio system **612** can further include a microphone for receiving audible signals of an end user. The audio system **612** can also be used for voice recognition applications. The UI **604** can further include an image sensor **613** such as a charged coupled device (CCD) camera for capturing still or moving images.

[0101] The power supply **614** can utilize common power management technologies such as replaceable and rechargeable batteries, supply regulation technologies, and/or charging system technologies for supplying energy to the components of the communication device **600** to facilitate long-range or short-range portable communications. Alternatively, or in combination, the charging system can utilize external power sources such as DC power supplied over a physical interface such as a USB port or other suitable tethering technologies.

[0102] The location receiver **616** can utilize location technology such as a global positioning system (GPS) receiver capable of assisted GPS for identifying a location of the communication device **600** based on signals generated by a constellation of GPS satellites, which can be used for facilitating location services such as navigation. The motion sensor **618** can utilize motion sensing technology such as an accelerometer, a gyroscope, or other suitable motion sensing technology to detect motion of the communication device **600** in three-dimensional space. The orientation sensor **620** can utilize orientation sensing technology such as a magnetometer to detect the orientation of the communication device **600** (north, south, west, and east, as well as combined orientations in

degrees, minutes, or other suitable orientation metrics).

[0103] The communication device **600** can use the transceiver **602** to also determine a proximity to a cellular, WiFi, Bluetooth®, or other wireless access points by sensing techniques such as utilizing a received signal strength indicator (RSSI) and/or signal time of arrival (TOA) or time of flight (TOF) measurements. The controller **606** can utilize computing technologies such as a microprocessor, a digital signal processor (DSP), programmable gate arrays, application specific integrated circuits, and/or a video processor with associated storage memory such as Flash, ROM, RAM, SRAM, DRAM or other storage technologies for executing computer instructions, controlling, and processing data supplied by the aforementioned components of the communication device **600**.

[0104] Other components not shown in FIG. **6** can be used in one or more embodiments of the subject disclosure. For instance, the communication device **600** can include a slot for adding or removing an identity module such as a Subscriber Identity Module (SIM) card or Universal Integrated Circuit Card (UICC). SIM or UICC cards can be used for identifying subscriber services, executing programs, storing subscriber data, and so on.

[0105] The terms “first,” “second,” “third,” and so forth, as used in the claims, unless otherwise clear by context, is for clarity only and doesn't otherwise indicate or imply any order in time. For instance, “a first determination,” “a second determination,” and “a third determination,” does not indicate or imply that the first determination is to be made before the second determination, or vice versa, etc.

[0106] In the subject specification, terms such as “store,” “storage,” “data store,” data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components described herein can be either volatile memory or nonvolatile memory, or can comprise both volatile and nonvolatile memory, by way of illustration, and not limitation, volatile memory, non-volatile memory, disk storage, and memory storage. Further, nonvolatile memory can be included in read only memory (ROM), programmable ROM (PROM), electrically programmable ROM (EPROM), electrically erasable ROM (EEPROM), or flash memory. Volatile memory can comprise random access memory (RAM), which acts as external cache memory. By way of illustration and not limitation, RAM is available in many forms such as synchronous RAM (SRAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), enhanced SDRAM (ESDRAM), Synchlink DRAM (SLDRAM), and direct Rambus RAM (DRRAM). Additionally, the disclosed memory components of systems or methods herein are intended to comprise, without being limited to comprising, these and any other suitable types of memory.

[0107] Moreover, it will be noted that the disclosed subject matter can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, mini-computing devices, mainframe computers, as well as personal computers, hand-held computing devices (e.g., PDA, phone, smartphone, watch, tablet computers, netbook computers, etc.), microprocessor-based or programmable consumer or industrial electronics, and the like. The illustrated aspects can also be practiced in distributed computing environments where tasks are performed by remote processing devices that are linked through a communications network; however, some if not all aspects of the subject disclosure can be practiced on stand-alone computers. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0108] In one or more embodiments, information regarding use of services can be generated including services being accessed, media consumption history, user preferences, and so forth. This information can be obtained by various methods including user input, detecting types of communications (e.g., video content vs. audio content), analysis of content streams, sampling, and

so forth. The generating, obtaining and/or monitoring of this information can be responsive to an authorization provided by the user. In one or more embodiments, an analysis of data can be subject to authorization from user(s) associated with the data, such as an opt-in, an opt-out, acknowledgement requirements, notifications, selective authorization based on types of data, and so forth.

[0109] Some of the embodiments described herein can also employ artificial intelligence (AI) to facilitate automating one or more features described herein. The embodiments (e.g., in connection with automatically identifying acquired cell sites that provide a maximum value/benefit after addition to an existing communication network) can employ various AI-based schemes for carrying out various embodiments thereof. Moreover, the classifier can be employed to determine a ranking or priority of each cell site of the acquired network. A classifier is a function that maps an input attribute vector, $X=(x_1, x_2, x_3, x_4, \dots, x_n)$, to a confidence that the input belongs to a class, that is, $f(x)=\text{confidence}(\text{class})$. Such classification can employ a probabilistic and/or statistical-based analysis (e.g., factoring into the analysis utilities and costs) to determine or infer an action that a user desires to be automatically performed. A support vector machine (SVM) is an example of a classifier that can be employed. The SVM operates by finding a hypersurface in the space of possible inputs, which the hypersurface attempts to split the triggering criteria from the non-triggering events. Intuitively, this makes the classification correct for testing data that is near, but not identical to training data. Other directed and undirected model classification approaches comprise, e.g., naïve Bayes, Bayesian networks, decision trees, neural networks, fuzzy logic models, and probabilistic classification models providing different patterns of independence can be employed. Classification as used herein also is inclusive of statistical regression that is utilized to develop models of priority.

[0110] As will be readily appreciated, one or more of the embodiments can employ classifiers that are explicitly trained (e.g., via a generic training data) as well as implicitly trained (e.g., via observing UE behavior, operator preferences, historical information, receiving extrinsic information). For example, SVMs can be configured via a learning or training phase within a classifier constructor and feature selection module. Thus, the classifier(s) can be used to automatically learn and perform a number of functions, including but not limited to determining according to predetermined criteria which of the acquired cell sites will benefit a maximum number of subscribers and/or which of the acquired cell sites will add minimum value to the existing communication network coverage, etc.

[0111] As used in some contexts in this application, in some embodiments, the terms “component,” “system” and the like are intended to refer to, or comprise, a computer-related entity or an entity related to an operational apparatus with one or more specific functionalities, wherein the entity can be either hardware, a combination of hardware and software, software, or software in execution. As an example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, computer-executable instructions, a program, and/or a computer. By way of illustration and not limitation, both an application running on a server and the server can be a component. One or more components may reside within a process and/or thread of execution and a component may be localized on one computer and/or distributed between two or more computers. In addition, these components can execute from various computer readable media having various data structures stored thereon. The components may communicate via local and/or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system, and/or across a network such as the Internet with other systems via the signal). As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry, which is operated by a software or firmware application executed by a processor, wherein the processor can be internal or external to the apparatus and executes at least a part of the software or firmware application. As yet another

example, a component can be an apparatus that provides specific functionality through electronic components without mechanical parts, the electronic components can comprise a processor therein to execute software or firmware that confers at least in part the functionality of the electronic components. While various components have been illustrated as separate components, it will be appreciated that multiple components can be implemented as a single component, or a single component can be implemented as multiple components, without departing from example embodiments.

[0112] Further, the various embodiments can be implemented as a method, apparatus or article of manufacture using standard programming and/or engineering techniques to produce software, firmware, hardware or any combination thereof to control a computer to implement the disclosed subject matter. The term “article of manufacture” as used herein is intended to encompass a computer program accessible from any computer-readable device or computer-readable storage/communications media. For example, computer readable storage media can include, but are not limited to, magnetic storage devices (e.g., hard disk, floppy disk, magnetic strips), optical disks (e.g., compact disk (CD), digital versatile disk (DVD)), smart cards, and flash memory devices (e.g., card, stick, key drive). Of course, those skilled in the art will recognize many modifications can be made to this configuration without departing from the scope or spirit of the various embodiments.

[0113] In addition, the words “example” and “exemplary” are used herein to mean serving as an instance or illustration. Any embodiment or design described herein as “example” or “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. Rather, use of the word example or exemplary is intended to present concepts in a concrete fashion. As used in this application, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or”. That is, unless specified otherwise or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

[0114] Moreover, terms such as “user equipment,” “mobile station,” “mobile,” subscriber station,” “access terminal,” “terminal,” “handset,” “mobile device” (and/or terms representing similar terminology) can refer to a wireless device utilized by a subscriber or user of a wireless communication service to receive or convey data, control, voice, video, sound, gaming or substantially any data-stream or signaling-stream. The foregoing terms are utilized interchangeably herein and with reference to the related drawings.

[0115] Furthermore, the terms “user,” “subscriber,” “customer,” “consumer” and the like are employed interchangeably throughout, unless context warrants particular distinctions among the terms. It should be appreciated that such terms can refer to human entities or automated components supported through artificial intelligence (e.g., a capacity to make inference based, at least, on complex mathematical formalisms), which can provide simulated vision, sound recognition and so forth.

[0116] As employed herein, the term “processor” can refer to substantially any computing processing unit or device comprising, but not limited to comprising, single-core processors; single-processors with software multithread execution capability; multi-core processors; multi-core processors with software multithread execution capability; multi-core processors with hardware multithread technology; parallel platforms; and parallel platforms with distributed shared memory. Additionally, a processor can refer to an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field programmable gate array (FPGA), a programmable logic controller (PLC), a complex programmable logic device (CPLD), a discrete gate or transistor logic, discrete hardware components or any combination thereof designed to

perform the functions described herein. Processors can exploit nano-scale architectures such as, but not limited to, molecular and quantum-dot based transistors, switches and gates, in order to optimize space usage or enhance performance of user equipment. A processor can also be implemented as a combination of computing processing units.

[0117] As used herein, terms such as “data storage,” “data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components or computer-readable storage media, described herein can be either volatile memory or nonvolatile memory or can include both volatile and nonvolatile memory.

[0118] What has been described above includes mere examples of various embodiments. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing these examples, but one of ordinary skill in the art can recognize that many further combinations and permutations of the present embodiments are possible.

Accordingly, the embodiments disclosed and/or claimed herein are intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims. Furthermore, to the extent that the term “includes” is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

[0119] In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be incorporated in or otherwise used in conjunction with other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue” indication reflects that the steps presented may be performed multiple times and/or may be succeeded by other activities not specifically shown. Further, while a flow diagram indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.

[0120] As may also be used herein, the term(s) “operably coupled to”, “coupled to”, and/or “coupling” includes direct coupling between items and/or indirect coupling between items via one or more intervening items. Such items and intervening items include, but are not limited to, junctions, communication paths, components, circuit elements, circuits, functional blocks, and/or devices. As an example of indirect coupling, a signal conveyed from a first item to a second item may be modified by one or more intervening items by modifying the form, nature or format of information in a signal, while one or more elements of the information in the signal are nevertheless conveyed in a manner that can be recognized by the second item. In a further example of indirect coupling, an action in a first item can cause a reaction on the second item, as a result of actions and/or reactions in one or more intervening items.

[0121] Although specific embodiments have been illustrated and described herein, it should be appreciated that any arrangement which achieves the same or similar purpose may be substituted for the embodiments described or shown by the subject disclosure. The subject disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, can be used in the subject disclosure. For instance, one or more features from one or more embodiments can be combined with one or more features of one or more other embodiments. In one or more embodiments, features that are positively recited can also be negatively recited and excluded from the embodiment with or without replacement by another structural and/or functional feature. The steps or functions described with respect to the embodiments of the subject disclosure can be performed in any order. The steps or functions described with respect to the embodiments of the subject disclosure can be performed alone or in combination with other steps or functions of the subject disclosure, as well as from other embodiments or from other steps that have not been

described in the subject disclosure. Further, more than or less than all of the features described with respect to an embodiment can also be utilized.

Claims

1. A device comprising: a remote user plane function (UPF) deployment and controller function (RUPDCF) for a plurality of UPFs of a communication network, wherein the RUPDCF is configured to communicate data for respective proprietary procedures for controlling respective UPFs of a plurality of UPFs; and a standard interface to facilitate communication between the RUPDCF and the plurality of UPFs, wherein the standard interface enables communication between a first UPF and a second UPF of the plurality of UPFs and a session management function (SMF), wherein the first UPF and the second UPF communicate with the SMF using proprietary extensions, wherein the RUPDCF facilitates communication between the first UPF and the second UPF and the SMF via the standard, service-based interface.
2. The device of claim 1, wherein the standard interface is implemented as a service-based interface (SBI).
3. The device of claim 1, wherein the RUPDCF comprises: a user data repository; and a configuration module configured to generate, update, and store configuration data for each UPF in the user data repository.
4. The device of claim 1, wherein the RUPDCF comprises: a testing module configured to perform simulated network testing and live network testing of UPF configurations.
5. The device of claim 1, wherein the standard interface further comprises: a secure communication module configured to enable mutual authentication between the RUPDCF and each UPF.
6. The device of claim 1, wherein the RUPDCF is configured to communicate data for controlling respective UPFs from multiple vendors according to vendor-proprietary control protocols.
7. The device of claim 6, wherein the RUPDCF is configured to translate the proprietary control protocols of the UPFs into a standardized protocol for communication with the SMF.
8. The device of claim 1, wherein the RUPDCF is configured to instruct a respective UPF of the plurality of UPFs to register with a network function repository to become discoverable by the SMF.
9. A method, comprising: providing, by a processing system including a processor, a controller function configured to operate independently of a user plane function (UPF) vendor and to orchestrate authentication, configuration, testing, and service state control of a UPF; establishing, by the processing system, a service-based interface between the controller function and the UPF to enable standardized communication; and automatically controlling, by the processing system, availability of the UPF in a communication network.
10. The method of claim 9, comprising: providing, by the processing system, the UPF with configuration data via the service-based interface.
11. The method of claim 10, wherein the providing the UPF with the configuration data comprises: providing, by the processing system, base configuration parameters and supplemental configuration information.
12. The method of claim 9, comprising: instructing, by the processing system, the UPF to perform an automated simulated test wherein performing the automated simulated test comprises establishing an association with a simulated session management function and exchanging simulated uplink and downlink traffic to verify session parameters and protocol rules.
13. The method of claim 9, comprising: performing, by the processing system, an automated live network test wherein the controller function is operative to direct a test user equipment to initiate a packet data unit session, establish a live association between a live session management function and the UPF via the service-based interface, and verify that the UPF correctly registers with a network repository function.

- 14.** The method of claim 13, comprising: instructing, by the processing system, the UPF to transition into an in-service state by registering with the network repository function using a designated service name, thereby making the UPF discoverable by session management functions, wherein the instructing is responsive to a successful live network test.
- 15.** A non-transitory machine-readable medium, comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations, the operations comprising: instantiating a remote user plane function (UPF) deployment and control function for a plurality of UPFs of a communication network, wherein respective UPFs of the plurality of UPFs utilize respective proprietary configuration and control interfaces, communicating data between the remote UPF deployment and control function for respective proprietary procedures and the plurality UPFs, wherein the communicating the data is via a standard, service-based interface; and controlling the plurality of UPFs by communicating via the standard, service-base interface.
- 16.** The non-transitory machine-readable medium of claim 15, wherein the controlling comprises: at least one of authentication, deployment, configuration, and testing a respective UPF of the plurality of UPFs, and wherein a first UPF and a second UPF of the plurality of UPFs are adapted to communicate with a session management function (SMF).
- 17.** The non-transitory machine-readable medium of claim 16, wherein the wherein the controlling further comprises: controlling availability of the respective UPFs of the plurality of UPFs, independent of a vendor of a respective UPF of the plurality of UPFs.
- 18.** The non-transitory machine-readable medium of claim 16, wherein the testing the respective UPF of the plurality of UPFs comprises: simulated testing and live network testing of a configuration of the plurality of UPFs.
- 19.** The non-transitory machine-readable medium of claim 15, wherein the operations further comprise: communicating data for the respective proprietary procedures for respective UPFs between respective UPFs of the plurality of UPFs and a session management function via the standard, service-based interface.
- 20.** The non-transitory machine-readable medium of claim 15, wherein the operations further comprise: receiving a registration of the plurality of UPFs at a network repository function (NRF) of the communication network to control availability of the plurality of UPFs to be placed in service.
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